

IDS Capstone

Capstone Group 45:

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Introduction

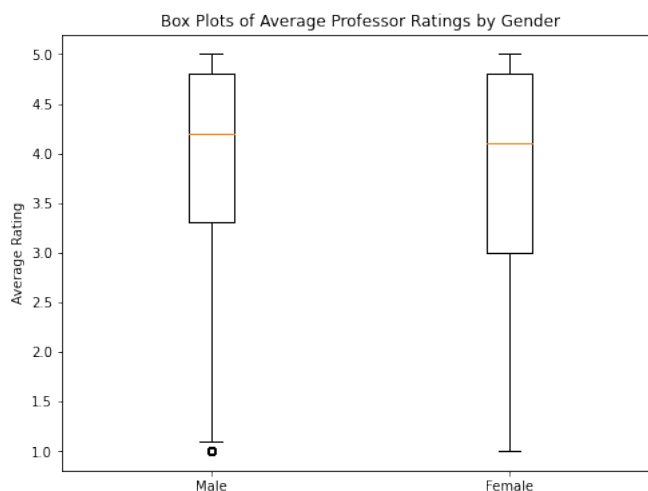
Author contributions: We met to discuss approaches to each question and then split up coding the tasks equally among the three members. Each person individually worked on their allotted tasks initially and then we met to discuss results and finalize our write-up.

Data pre-processing: There are 77733 NaN values in the column ‘Proportion of students that would take the course again’ and 19889 NaN values in all other columns of rmpCapstoneNum.csv. Our approach to these values was to delete all rows where the average rating was NaN for all parts of this project. For the remaining 57844 NaN values, we attempted an approach with mean imputation for that column and a second approach with removal of all NaN values for Q6. For the remaining linear regression tasks, we decided to remove NaN values as mean imputation was not achieving good performance. For Q10, the classification task, we still used mean imputation as we were able to achieve fairly good results.

Random seed: We seeded all random number generators (train_test_split as well as rng for bootstrapping and downsampling) with Rebecca’s N number: N16115289

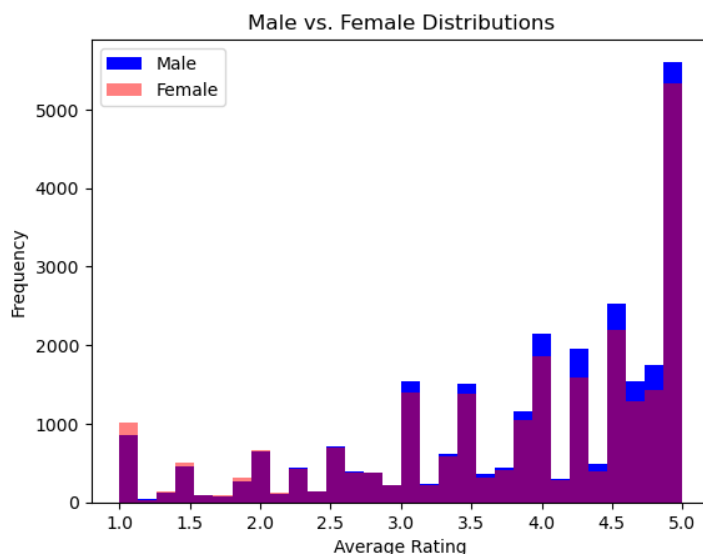
1. Evidence of Gender Bias in Ratings

To determine if there is evidence of a pro-male gender bias in average rating, we performed a one-tailed Mann-Whitney U test under the null hypothesis that there is no difference in ratings between male and female professors. The p-value was 4.27×10^{-6} which is far lower than our alpha threshold of 0.005. Therefore, we drop the null hypothesis, and conclude that there is evidence of a pro-male average rating bias in the data set. Although our analysis benefits from a large sample size and avoid p-hacking, it is not possible to fully account for all potential confounding variables without running an experiment. As a result, critics may still find groups to question the findings.



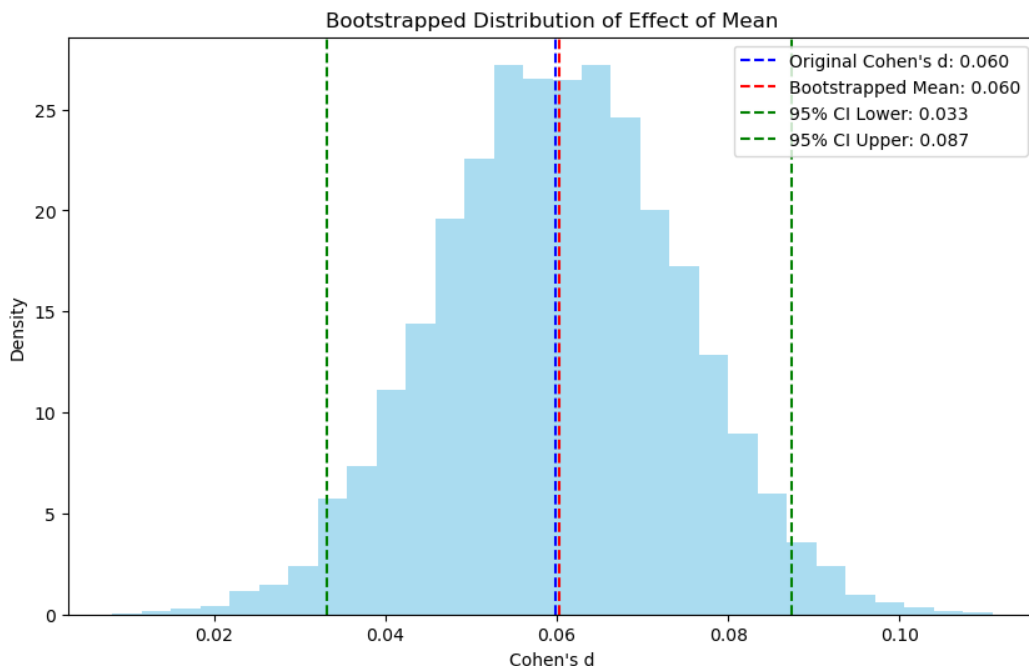
2. Gender Differences in Variance of Ratings

We compared the two distributions using the K-S test to test the null hypothesis that variance doesn't differ between the two groups. The K-S test compares whether data come from the same distribution. Similar distributions would have similar variance, so this test indirectly compares similarity of variance. The p-value was 3.92×10^{-8} and the statistic was 0.026. The two histograms below plot the average rating distribution for male and female professors. Given that the p-value was so low (< 0.005), the difference in the variance of the average ratings for male and female professors is significant.



3. Size of Gender Effects on Ratings

We used bootstrapping to estimate the effect sizes of the difference in means and the difference in variances using Cohen's d as a measure of effect size. Bootstrapping was used because the data does not distribute asymptotically normally. To find the effect size of the difference in variance between the male and female groups, we replaced the sample means in the equation with the sample variances. The effect size of gender bias in average rating is 0.06, with a 95% confidence interval of [0.033, 0.087]. The effect size of gender bias in spread of average rating is 0.0529, with a 95% confidence interval of [0.033, 0.072]. These effect sizes show that with the size of our dataset, we should be able to capture the difference in average rating and spread of average rating, which is the case as we found significant differences in both.



4. Gender Differences in Tags Awarded

To determine whether there are gender differences in the tags awarded to professors, we first normalized the tag counts. Normalization involved dividing the raw count of each tag by the number of ratings each professor received. This ensures fairness by accounting for the fact that professors with more reviews naturally tend to receive more tags.

We then performed a two-tailed t-test for each tag to determine if there was a statistically significant difference in how tags were assigned to male versus female professors. The t-test identifies whether the average values of two groups (male and female) are significantly different.

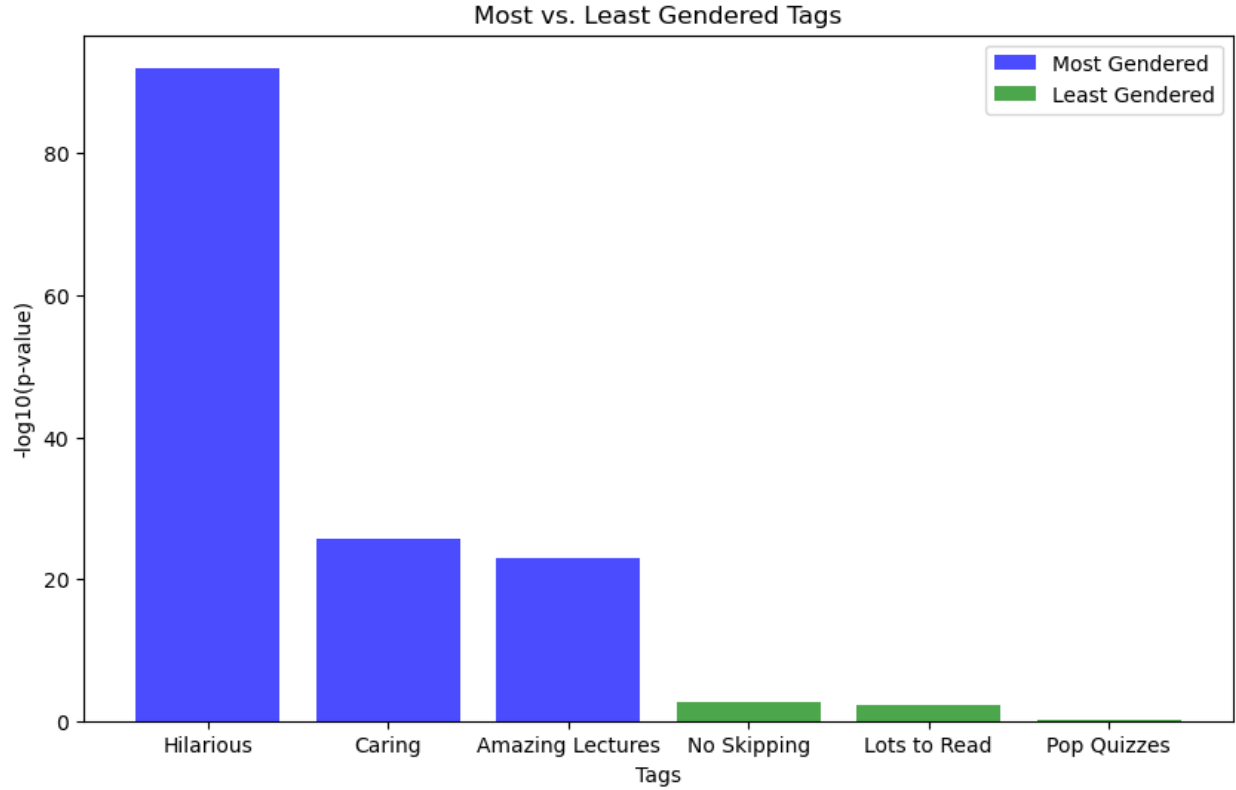
This method was chosen because normalization removes biases caused by differing numbers of reviews, and the t-test provides a robust way to assess statistical differences between groups.

The results showed:

- The most gendered tags (those with the smallest p-values, indicating strong evidence of a difference) were:
 1. Hilarious ($p \approx 8.90 \times 10^{-93}$).
 2. Caring ($p \approx 1.67 \times 10^{-26}$).
 3. Amazing Lectures ($p \approx 1.39 \times 10^{-23}$).
- The least gendered tags (those with the largest p-values, indicating minimal evidence of a difference) were:

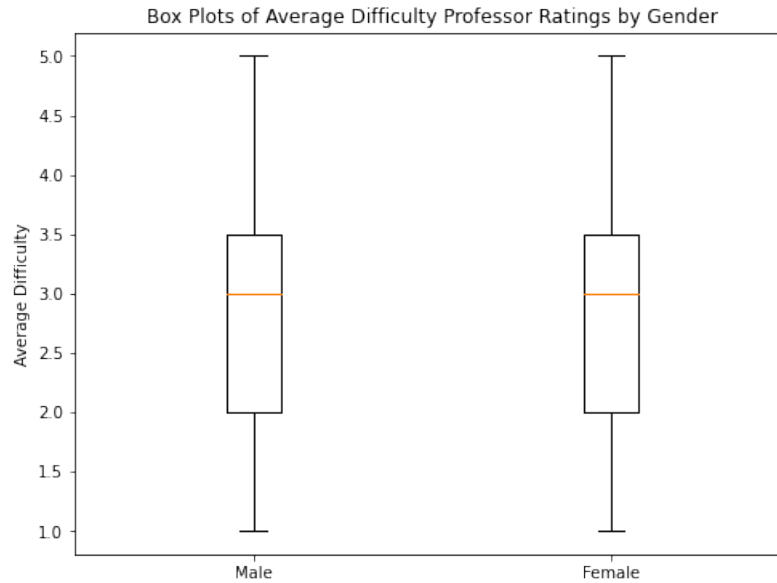
1. No Skipping ($p \approx 0.7056$).
2. Lots to Read ($p \approx 0.0050$).
3. Pop Quizzes ($p \approx 0.0023$).

The figure below summarizes the results, with bars representing the statistical significance ($-\log_{10}(p)$) of the most and least gendered tags.



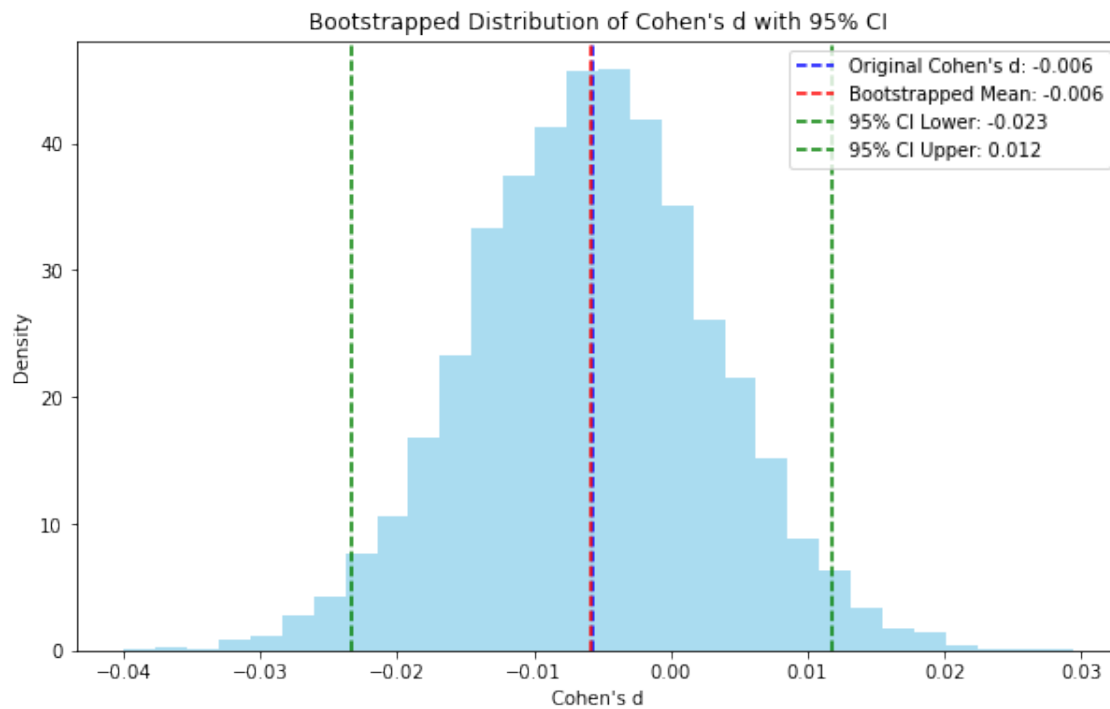
5. Gender Differences in Perceived Difficulty

To determine the presence or absence of a gender difference in terms of average difficulty, we conducted a two-tailed Mann-Whitney U test under the null hypothesis that there is no gender difference in average difficulty. We found a p-value of 0.678, which is greater than 0.005, so we maintained our null hypothesis and concluded that this difference is not statistically significant.



6. Size of Gender Effects on Difficulty

To quantify the likely size of this effect at 95% confidence, we calculated the effect size between how males versus females perceive difficulty using Cohen's d . Because the data doesn't distribute asymptotically normally, we also used bootstrapping to repeatedly resample the two groups. We found the effect size to be -0.006 with a 95% confidence interval of $[-0.023, 0.012]$. These results suggest that there is almost no difference in the means of the two groups relative to their pooled standard deviation. The negative sign suggests that males on average have slightly lower values; however, this difference is ultimately negligible. Because the confidence interval contains zero, this indicates that there is no statistically significant difference between the two groups at the 95% confidence level.



7. Regression Model for Average Ratings Based on Numerical Factors

To predict average ratings of professors, we used two distinct approaches to prepare the data. The predictors were numerical columns, and the target variable was the average rating. We dropped all professors whose average rating was NaN.

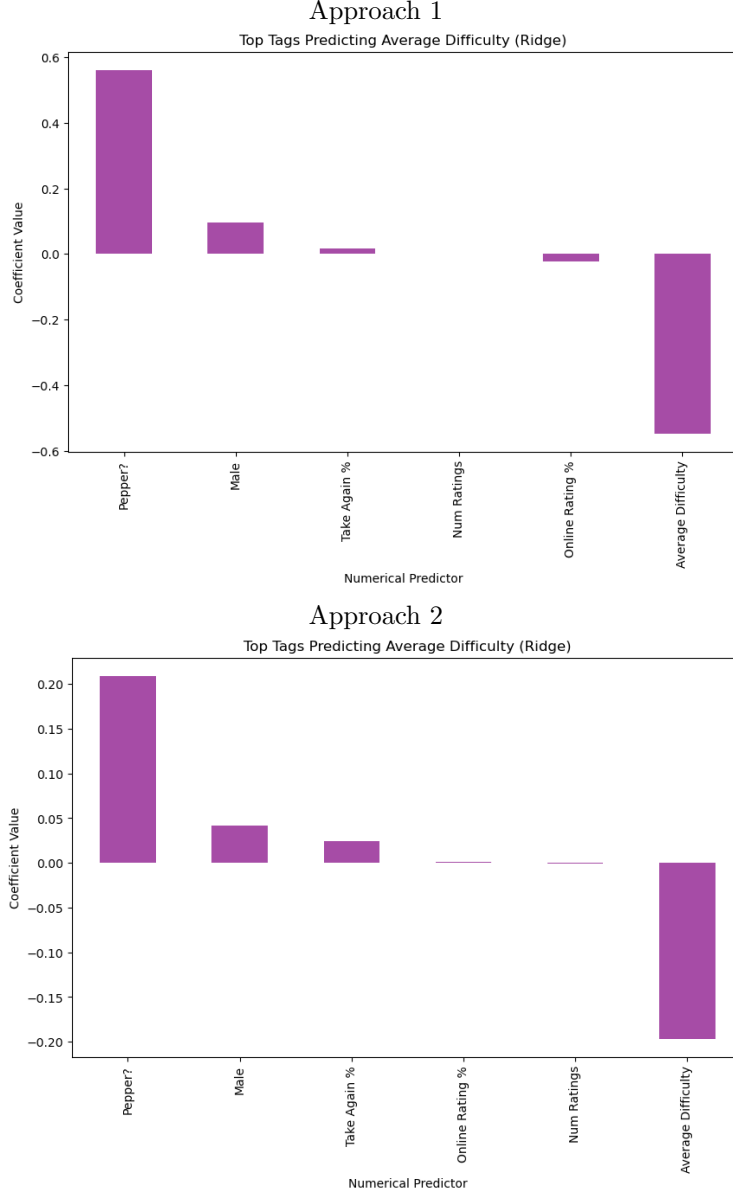
To address collinearity, we removed the "Female" column, as it is highly collinear with the "Male" column. By calculating the Variance Inflation Factor (VIF) for each predictor, we determined that the "Take Again Proportion" column was introducing collinearity because it had a VIF greater than 10. However, removing this column negatively affected the predictive capabilities of the model.

Approach 1: Impute the mean for the proportion of students that would take the course again.

Approach 2: Remove all NaN values and work with a much smaller dataset.

For both approaches, we split the dataset into 80% training and 20% testing, and used Ridge regression to obtain the following results:

	Approach 1	Approach 2
R^2	0.376	0.809
RMSE	0.890	0.376



The most predictive features in both cases were receiving a pepper, gender, and average difficulty rating for the course. Approach 2 is problematic; while we found a much higher R^2 value, meaning the model explains more of the variability observed in average ratings, we were left with a dramatically smaller sample size of only 12,160 professors. Additionally, the presence of a value in the column of proportion of students that would take the course again was positively correlated with average rating.

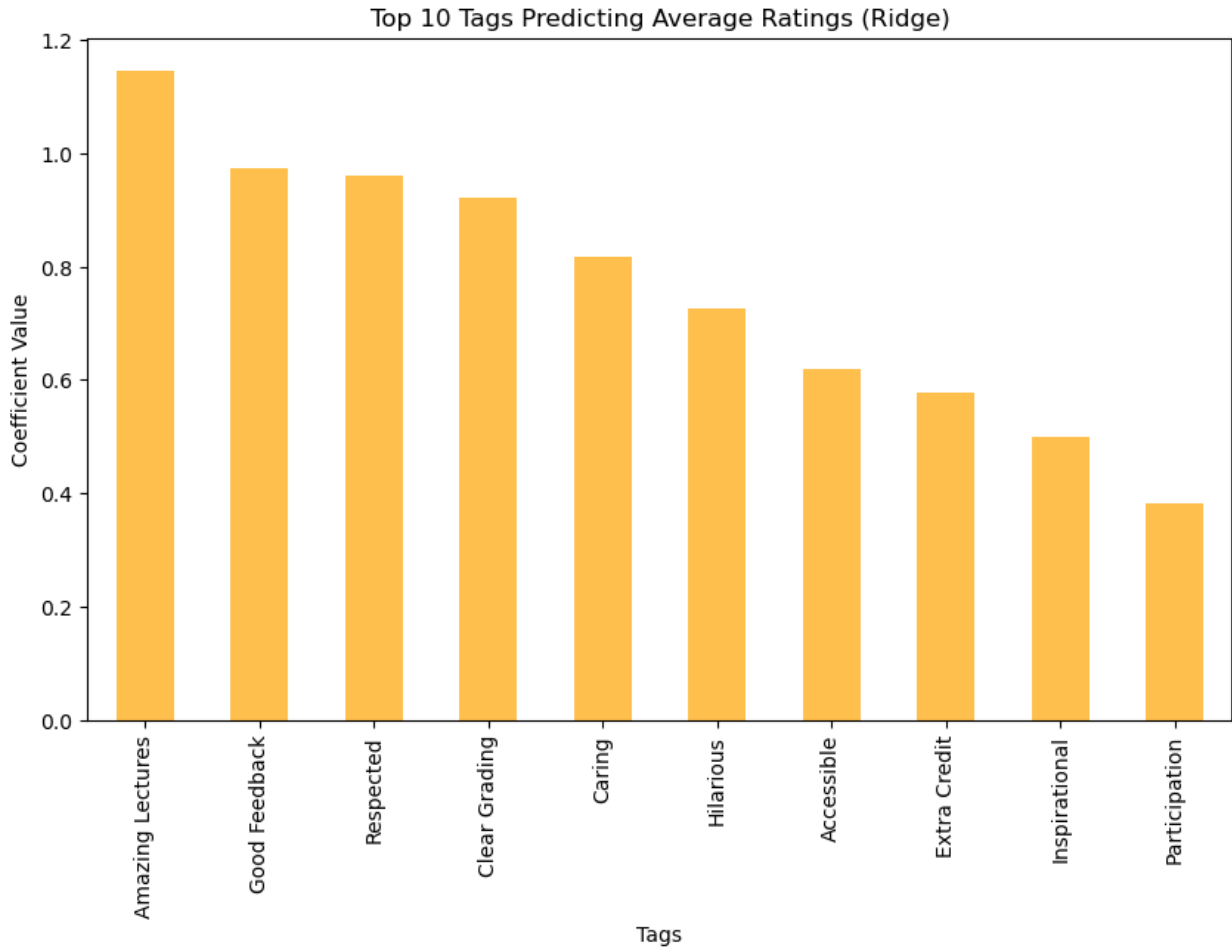
8. Regression Model for Average Ratings Based on Tags

To predict professors' average ratings, we developed a regression model using the normalized tag counts as predictors. To address potential collinearity between tags, we again used VIF and removed predictors with a VIF greater than 10 as they indicate significant multicollinearity. Next, we applied Ridge regression, which introduces regularization to stabilize coefficients and mitigate the effects of remaining collinearity. The dataset was split into 80% for training and 20% for testing. Ridge regression was chosen because it maintains predictive performance while reducing the influence of collinear predictors.

Results:

- $R^2 = 0.752$: The model explains 75.2% of the variability in ratings.
- RMSE = 0.429: The average prediction error is approximately 0.429 points.
- The most predictive tags were:
 1. Amazing Lectures (Coefficient = 1.146).
 2. Good Feedback (Coefficient = 0.974).
 3. Respected (Coefficient = 0.960).

The bar chart below shows the top 10 tags most predictive of average ratings after addressing collinearity. Tags with higher coefficients have a stronger positive influence on ratings.



9. Regression Model for Average Difficulty Based on Tags

To predict the perceived difficulty of professors, we used the same Ridge regression approach as in Question 8. The predictors were normalized tag counts, and the target variable was the average difficulty rating. Again, we addressed collinearity by removing predictors with high VIF values and using Ridge regression for regularization.

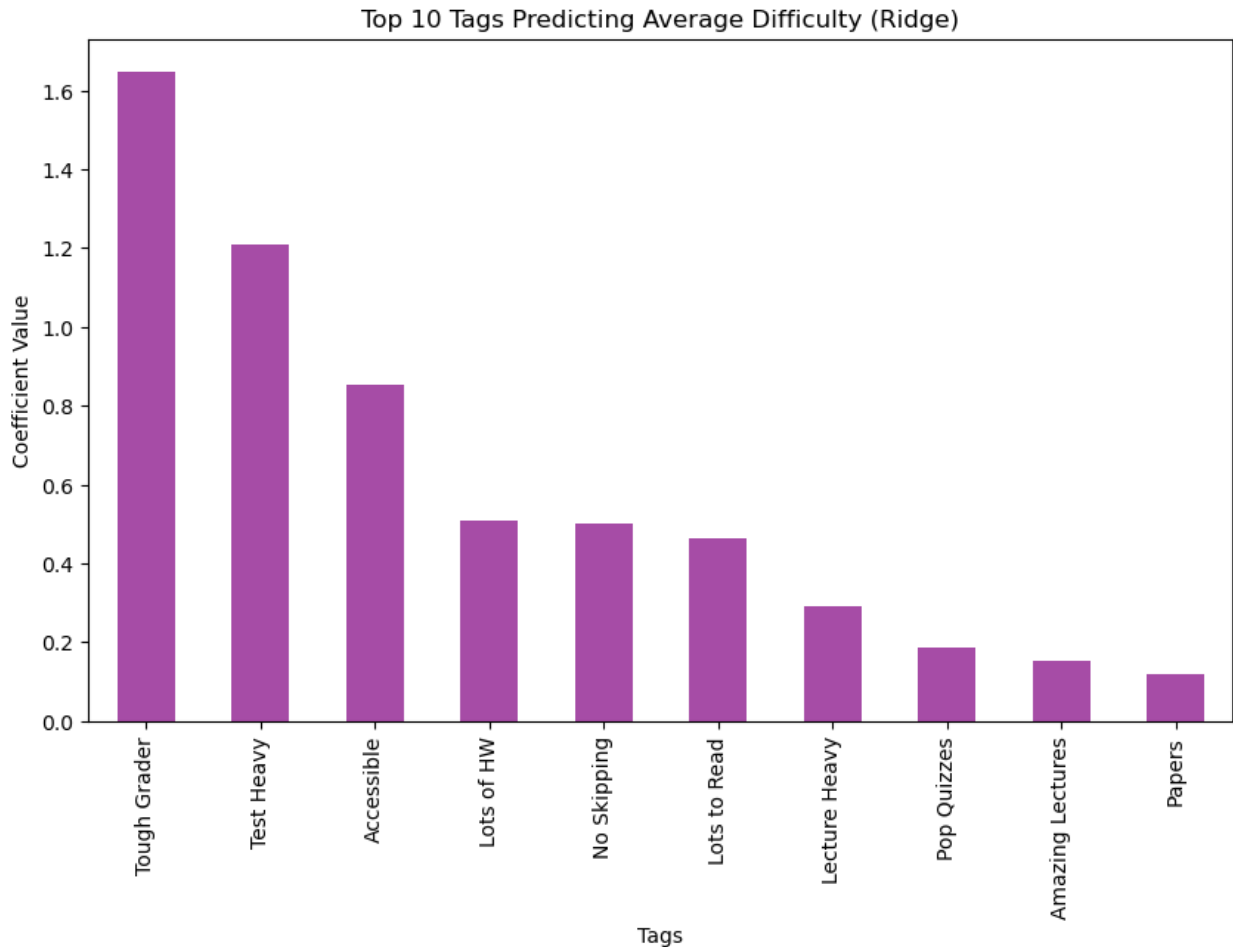
The dataset was split into 80% training and 20% testing. This ensured that the model's performance was evaluated on unseen data, improving its robustness.

Results:

- $R^2 = 0.556$: The model explains 55.6% of the variability in difficulty ratings.

- RMSE = 0.480: The average prediction error is approximately 0.480 points.
- The most predictive tags were:
 1. Tough Grader (Coefficient = 1.648).
 2. Test Heavy (Coefficient = 1.208).
 3. Accessible (Coefficient = 0.854).

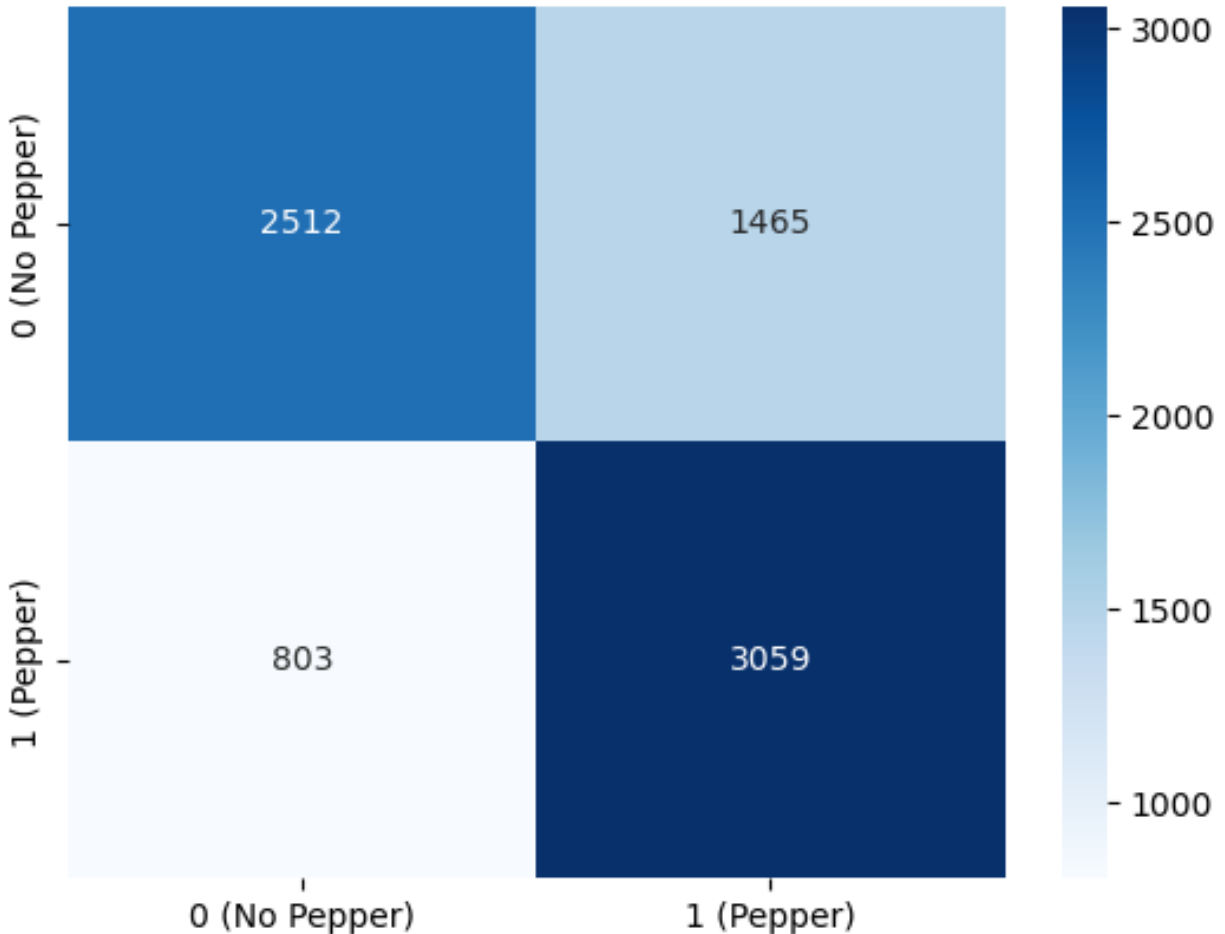
The bar chart below displays the top 10 tags most predictive of difficulty ratings after addressing collinearity. Tags with higher coefficients are more strongly associated with increased difficulty.



10. Classification Model for Receiving a Pepper

To handle the remaining NaN values in the column 'Take Again %', we imputed the mean value of the column. We regularized the tags data by the number of reviews received as the number of tags is dependent on the number of reviews received. To address class imbalance concerns, we downsampled the larger group (professors who received 'No Pepper') so that the classes would be the same size. We used a logistic regression classification model to classify the test data (20% of the data) into the two groups. Using the true positive and false positive rate, we set a threshold of 0.465. AUC was 0.77, which means the model has a fair ability to discriminate between the two classes. Potential improvements could include using a non-linear model like a random forest to better discriminate between classes. Below are the results of the logistic classifier:

	precision	recall	f1-score	support
0.0	0.76	0.63	0.69	3977
1.0	0.68	0.79	0.73	3862
accuracy			0.71	7839
macro avg	0.72	0.71	0.71	7839
weighted avg	0.72	0.71	0.71	7839



11 Extra Credit

We split the professors by field (stem vs. non-stem fields) using keywords. We removed all professors whose field has ≤ 15 participants (i.e. they are in a niche field). Then we conducted a one-tailed Mann-Whitney U-test to determine if professors teaching in STEM fields were rated higher, and if they were rated more difficult. A one-tailed test was conducted because we are interested in if STEM professors had a specifically higher rating.

Results:

Higher Average Rating: Our p-value was 0.467.

Higher Difficulty: Our p-value was 0.355.

Given that both p-values were high, it is reasonable to conclude that there is no significant difference in how students rate professors in STEM or non-STEM fields. This can be seen in the histograms below which show similar distributions for both groups. One thing to note is that there is at least one confounder: students who find STEM fields difficult and vice versa are not likely to take courses with those courses.

